



SQL PIZZA PROJECT

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I'm a Data Analytics learner with 24 years of cross-industry experience in pharma sales, mobile handsets, and education. Two months ago, I began a focused transition into analytics, consistently upskilling in Excel, SQL, Python, Power BI, Tableau, and Statistics. I've completed 8 domain-specific projects during a virtual internship, applying insights to real-world business challenges. My background in communication and training enhances my ability to translate complex data into clear, actionable strategies. I'm now actively building a future in data-driven decision-making.





What is the project

We have 4 different csv files containing details of various pizza types and the order details



Name	Date modified	Type	Size
 order_details	3/6/2024 4:13 PM	Microsoft Excel Co...	1,278 KB
 orders	3/6/2024 4:13 PM	Microsoft Excel Co...	553 KB
 pizza_types	3/6/2024 4:13 PM	Microsoft Excel Co...	4 KB
 pizzas	3/6/2024 4:13 PM	Microsoft Excel Co...	4 KB



What is the project

We need to work on the files creating database and tables to work on the tasks as stated in the next slide





Project Tasks

Level 1

- Retrieve the total number of orders placed.
- Calculate the total revenue generated from pizza sales.
- Identify the highest-priced pizza.
- Identify the most common pizza size ordered.
- List the top 5 most ordered pizza types along with their quantities.

Level 2

- Join the necessary tables to find the total quantity of each pizza category ordered.
- Determine the distribution of orders by hour of the day.
- Join relevant tables to find the category-wise distribution of pizzas.
- Group the orders by date and calculate the average number of pizzas ordered per day.
- Determine the top 3 most ordered pizza types based on revenue.

Level 3

- Calculate the percentage contribution of each pizza type to total revenue.
- Analyze the cumulative revenue generated over time.
- Determine the top 3 most ordered pizza types based on revenue for each pizza category.





Data Preparation

Creating database and loading the data from csv files

```
create database if not exists pizzastore;

use pizzastore;

select * from pizzas;

load data infile "F:/SQL Project/pizza_sales/order_details.csv"
into table order_details
fields terminated by ","
enclosed by '"'
lines terminated by "\n"
ignore 1 rows
(order_details_id, order_id, pizza_id, quantity);
```

```
load data infile "F:/SQL Project/pizza_sales/orders.csv"
into table orders
fields terminated by ","
enclosed by '"'
lines terminated by "\n"
ignore 1 rows
(order_id, date, time);
```





Level 1 - Task 1



```
-- Task Level 1
-- 1. Retrieve the total number of orders placed.

select count(order_id) as Total_Orders
from orders;

-- Date wise orders

select date, count(*) As Date_wise_orders
from orders
group by date
order by date;
```

Result Grid			Filter Rows
Total_Orders			
▶	21384		

Result Grid			Filter Rows:
	date	Date_wise_orders	
▶	2015-01-01	103	
	2015-01-02	67	
	2015-01-03	66	
	2015-01-04	52	
	2015-01-05	54	
	2015-01-06	64	



Level 1 - Task 2 - 1



-- 2. Calculate the total revenue (pizza type wise) generated from pizza sales.

```
select o.pizza_id,  
sum(o.quantity) as Total_Quantity,  
sum(p.price * o.quantity) As Revenue  
from order_details o  
join pizzas p  
on p.pizza_id = o.pizza_id  
group by o.pizza_id;
```

Result Grid			
Filter Rows:			
	pizza_id	Total_Quantity	Revenue
▶	bbq_ckn_s	519	6617.25
	bbq_ckn_m	1022	17118.5
	bbq_ckn_l	1068	22161
	cali_ckn_s	539	6872.25
	cali_ckn_m	1024	17152
	cali_ckn_l	982	20376.5



Level 1 - Task 2 - 2



```
-- 2. Calculate the total revenue generated from pizza sales.
```

```
with rev as (  
  select o.pizza_id,  
         sum(o.quantity) as Total_Quantity,  
         sum(p.price * o.quantity) As Revenue  
  from order_details o  
  join pizzas p  
  on p.pizza_id = o.pizza_id  
  group by o.pizza_id  
  order by Revenue desc)
```

```
select sum(Revenue) as Total_Revenue  
from rev;
```

Result Grid  Filter R	
	Total_Revenue
▶	878626.000000000005



Level 1 - Task 3



-- 3. Identify the highest-priced pizza.

```
select pt.name, p.price
from pizza_types pt
join pizzas p
on pt.pizza_type_id = p.pizza_type_id
order by p.price desc limit 3;
```

Result Grid			Filter Rows:
	name	price	
▶	The Greek Pizza	35.95	
	The Greek Pizza	25.5	
	The Brie Carre Pizza	23.65	



Level 1 - Task 4

-- 4. Identify the most common pizza size ordered.

```
SELECT
    p.size, COUNT(o.order_details_id) AS Ordered_Qty
FROM
    pizzas p
    JOIN
    order_details o ON p.pizza_id = o.pizza_id
GROUP BY p.size
ORDER BY Ordered_Qty DESC;
```

Result Grid			Filter R
	size	Ordered_Qty	
▶	L	19914	
	M	16503	
	S	15213	
	XL	586	
	XXL	30	





Level 1 - Task 5



```
-- 5. List the top 5 most ordered pizza types along with their quantities.

select pt.name as Pizza_Name, sum(o.quantity) as Total_Qty
from pizzas p
join pizza_types pt
on p.pizza_type_id = pt.pizza_type_id
join order_details o
on p.pizza_id = o.pizza_id
group by pt.name
order by Total_Qty desc limit 5;
```

Result Grid			Filter Rows:
	Pizza_Name	Total_Qty	
▶	The Pepperoni Pizza	2633	
	The Classic Deluxe Pizza	2615	
	The Barbecue Chicken Pizza	2609	
	The Hawaiian Pizza	2584	
	The Thai Chicken Pizza	2551	



Level 2 - Task 1



```
-- 1. Join the necessary tables to find the total quantity of each pizza category ordered.

select pt.category as Pizza_Category, sum(o.quantity) as Total_Qty
from pizzas p
join pizza_types pt
on p.pizza_type_id = pt.pizza_type_id
join order_details o
on p.pizza_id = o.pizza_id
group by pt.category
order by Total_Qty desc limit 5;
```

Result Grid			Filter Rows:
	Pizza_Category	Total_Qty	
▶	Classic	15987	
	Supreme	12891	
	Veggie	12540	
	Chicken	11850	



Level 2 - Task 2 - 1



-- 2. Determine the distribution of orders by hour of the day.

```
select hour(time) as Hours, count(order_id) As Total_Orders
from orders
group by Hours;
```

Result Grid			Filter Row
	Hours	Total_Orders	
▶	11	1233	
	12	2527	
	13	2465	
	14	1479	
	15	1475	
	16	1921	
	17	2336	
	18	2399	
	19	2009	
	20	1642	
	21	1198	
	22	663	
	23	28	
	10	8	
	9	1	



Level 2 - Task 2 - 2



```
-- which hour of the day observes the maximum orders

select hour(time) as Hours, count(order_id) As Total_Orders
from orders
group by Hours
order by Total_Orders desc;

-- Maximum orders are received at 12:00 PM followed by 1:00 PM and 6:00 PM
```

Result Grid					Filter
	Hours	Total_Orders			
▶	12	2527			
	13	2465			
	18	2399			
	17	2336			
	19	2009			
	16	1921			
	20	1642			
	14	1479			
	15	1475			
	11	1233			
	21	1198			
	22	663			
	23	28			
	10	8			
	9	1			



Level 2 - Task 3



```
-- 3. Join relevant tables to find the category-wise distribution of pizzas.
```

```
select category, count(pizza_type_id) as Total  
from pizza_types  
group by category;
```

Result Grid			Filter
	category	Total	
▶	Chicken	6	
	Classic	8	
	Supreme	9	
	Veggie	9	



Level 2 - Task 4



```
-- 4. Group the orders by date and calculate the average number of pizzas ordered per day.

select avg(Total_Orders)
from (select o.date as Date, sum(od.quantity) as Total_Orders
from orders o
join order_details od
on o.order_id = od.order_id
group by o.date) as Ord;
```

Result Grid			Filter Rows:
	Date	Total_Orders	
▶	2015-01-01	496	
	2015-01-02	330	
	2015-01-03	316	
	2015-01-04	212	
	2015-01-05	250	
	2015-01-06	294	
	2015-01-07	276	
	2015-01-08	346	
	2015-01-09	254	
	2015-01-10	292	

Result Grid		Filter Rows:
	avg(Total_Orders)	
▶	149.2737	



Level 2 - Task 5



-- 5. Determine the top 3 most ordered pizza types based on revenue.

```
select pt.name as Name, round(sum(p.price * od.quantity),0) As Revenue
from pizzas p
join order_details od
on p.pizza_id = od.pizza_id
join pizza_types pt
on p.pizza_type_id = pt.pizza_type_id
group by pt.name
order by Revenue desc limit 3;
```

Result Grid			Filter Rows:
	Name	Revenue	
▶	The Thai Chicken Pizza	46733	
	The Barbecue Chicken Pizza	45897	
	The California Chicken Pizza	44401	



Level 3 - Task 1



```
-- 1. Calculate the percentage contribution of each pizza type to total revenue.

select pt.category as Category,
round(sum(p.price * od.quantity),0) as Revenue,
round(sum(p.price * od.quantity) /
(select sum(p.price * od.quantity)
from pizzas p join order_details od on p.pizza_id = od.pizza_id) * 100,2)
as Contribution
from pizzas p
join order_details od
on p.pizza_id = od.pizza_id
join pizza_types pt
on p.pizza_type_id = pt.pizza_type_id
group by pt.category
order by Revenue desc;
```

Result Grid   Filter Rows:			
	Category	Revenue	Contribution
▶	Classic	220053	26.91
	Supreme	208197	25.46
	Chicken	195920	23.96
	Veggie	193690	23.68



Level 3 - Task 2



```
-- 2. Analyze the cumulative revenue generated over time.

select o.date as Date, round(sum(p.price * od.quantity),0) as Revenue,
round(sum(sum(p.price * od.quantity)) over(order by o.date), 0) as Cum_Rev
from orders o
join order_details od
on o.order_id = od.order_id
join pizzas p
on od.pizza_id = p.pizza_id
group by o.date;
```

Result Grid   Filter Rows:			
	Date	Revenue	Cum_Rev
▶	2015-01-01	2714	2714
	2015-01-02	2732	5446
	2015-01-03	2662	8108
	2015-01-04	1755	9864
	2015-01-05	2066	11930
	2015-01-06	2429	14358
	2015-01-07	2202	16561
	2015-01-08	2838	19399
	2015-01-09	2127	21526
	2015-01-10	2464	23990
	2015-01-11	1872	25863
	2015-01-12	1919	27782
	2015-01-13	2050	29831



Level 3 - Task 3



```
-- 3. Determine the top 3 most ordered pizza types based on revenue for each pizza category.

select category, Revenue, ranked
from(
  -- creating a CTE for getting the categories and the corresponding revenues
  (with Ranking as (
    select pt.category as category, pt.name, round(sum(p.price * od.quantity),0) as Revenue
    from pizza_types pt
    join pizzas p
    on pt.pizza_type_id = p.pizza_type_id
    join order_details od
    on od.pizza_id = p.pizza_id
    group by pt.category, pt.name
    order by Revenue desc)
    -- Ranking based on revenue for each category
    select category, Revenue,
    rank() over(partition by category order by Revenue desc) as ranked
    from Ranking)) as temp
  -- limiting the ranking up to 3
  where ranked <=3;
```

Result Grid				Filter Rows:
	category	Revenue	ranked	
▶	Chicken	43434	1	
	Chicken	42768	2	
	Chicken	41410	3	
	Classic	38180	1	
	Classic	32273	2	
	Classic	30162	3	
	Supreme	34831	1	
	Supreme	33477	2	
	Supreme	30940	3	
	Veggie	32266	1	
	Veggie	26781	2	
	Veggie	26066	3	



Analysis Areas Covered

- **Total Revenue, Quantity Sold, Average Order Value**
- **Datewise Sales Trends**
- **Sales by Category, Size, Day, and Hour**
- **Top Selling Pizzas**
- **Cumulative Revenue with WINDOW FUNCTION**

THANKS FOR YOUR ATTENTION

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